Problem 1: Higher-Order Short-Time Hypocoercivity

Human-Readable Research Log

Problem ID: `problem-1-higher-order-short-time`

Status: Part (a) solved; part (b) reduced to a complete analytic spectral-jet construction with exact coefficient formulas.

Setting: Finite-dimensional complex matrices and the Euclidean operator norm.

Contents

1 Operational specification

1.1 Objective

For

$$P(t) = e^{-Ct}, \qquad C_H := \frac{C + C^*}{2} \geq 0,$$

let

$$f(t) := \|P(t)\|_2^2.$$

The primary hypocoercivity index m gives

$$f(t) = 1 - c_1 t^{2m+1} + O(t^{2m+2}).$$

The tasks are:

1. determine the coefficient at order $2m + 2$, and decide whether it requires a new integer index;
2. give a mathematically complete mechanism that determines every subsequent Taylor coefficient, including degenerate singular-value branches.

1.2 Success criterion

A successful result must:

- prove the claimed coefficient identities, not infer them only from examples;
- distinguish the coercive edge case $m = 0$ from the genuinely hypocoercive case $m \geq 1$;
- handle multiplicities of the maximal singular-value branch;
- provide an exact finite-order algorithm for all higher coefficients;
- keep the coefficient convention for the norm separate from that for the squared norm.

1.3 Non-counting outcomes and known failure modes

The following would not solve the problem:

- expanding one fixed trajectory instead of the propagator norm envelope;
- assuming the largest singular value is simple at $t = 0$;
- confusing the coefficient for $\|P(t)\|_2$ with that for $\|P(t)\|_2^2$;
- claiming $f(-t) = 1/f(t)$ (this is false in general);
- postulating integer indices that cannot encode continuously varying coefficients;
- omitting the coercive case $m = 0$.

2 Known leading term

Let

$$K_{m-1} := \ker\left(\sum_{j=0}^{m-1} (C^*)^j C_H C^j\right).$$

For $m \geq 1$, Theorem 2.7 of Achleitner–Arnold–Carlen gives the leading coefficient of the norm. Therefore the leading coefficient of the **squared** norm is

$$\boxed{c_1 = \frac{2}{(2m+1)!\binom{2m}{m}} \min_{\substack{x \in K_{m-1} \\ \|x\|=1}} \langle x, (C^*)^m C_H C^m x \rangle.}$$

For $m = 0$,

$$\boxed{c_1 = 2\lambda_{\min}(C_H).}$$

Reference: F. Achleitner, A. Arnold, E. A. Carlen, *The hypocoercivity index for the short time behavior of linear time-invariant ODE systems*, J. Differential Equations 371 (2023), 83–115, Theorem 2.7.

3 Main structural theorem: an odd logarithmic normal form

3.1 Theorem

For every matrix C, there is an analytic Hermitian matrix function $A_C(s)$ near $s = 0$ and an analytic eigenvalue branch $\alpha_*(s)$ of $A_C(s)$, maximal for all sufficiently small $s > 0$, such that

$$\boxed{\log \|e^{-Ct}\|_2^2 = t\,\alpha_*(t^2), \qquad 0 < t < \varepsilon.}$$

Consequently, the right-hand Taylor expansion of the logarithmic squared norm contains only odd powers:

$$\boxed{\log f(t) = \sum_{r=0}^{\infty} \kappa_r t^{2r+1}, \qquad \alpha_*(s) = \sum_{r=0}^{\infty} \kappa_r s^r.}$$

This is stronger than the parity of the leading exponent: it constrains the entire local expansion.

3.2 Canonical construction

Set

$$Q(t) := e^{-C^*t}e^{-Ct}, \qquad Z(t) := \operatorname{Log} Q(t),$$

where Log is the principal matrix logarithm near the identity. Also set

$$S(t) := e^{C^*t} = U(t)R(t)$$

to be the right polar decomposition, with $U(t)$ unitary and $R(t) > 0$. Let

$$W(t) := U(t)^{-1/2}$$

use the principal unitary square root near the identity, and define

$$\widehat{Z}(t) := W(t)^* Z(t) W(t).$$

Then $\widehat{Z}$ is analytic, Hermitian, and exactly odd. Hence there is a unique analytic Hermitian $A_C(s)$ such that

$$\widehat{Z}(t) = tA_C(t^2).$$

Because $\widehat{Z}(t)$ and $Z(t)$ are unitarily similar, they have the same eigenvalues.

3.3 Proof

First,

$$Q(-t) = e^{C^*t}e^{Ct} = S(t)Q(t)^{-1}S(t)^{-1}.$$

Functional calculus therefore gives

$$Z(-t) = -S(t)Z(t)S(t)^{-1}. \tag{3.1}$$

Use the following elementary lemma.

Hermitian-similarity lemma. If Hermitian matrices A and B satisfy $B = SAS^{-1}$ and $S = UR$ is the right polar decomposition, then R commutes with A and $B = UAU^*$.

Indeed, Hermiticity of RAR^{-1} gives

$$RAR^{-1} = R^{-1}AR,$$

so $R^2A = AR^2$; positivity of R then implies $RA = AR$.

Apply the lemma to $A = -Z(t)$ and $B = Z(-t)$ in (3.1). It follows that

$$Z(-t) = -U(t)Z(t)U(t)^*. \tag{3.2}$$

The polar unitary of $S(t)^{-1} = S(-t)$ is $U(t)^*$, so

$$U(-t) = U(t)^*.$$

Therefore

$$W(-t) = U(-t)^{-1/2} = U(t)^{1/2} = U(t)W(t).$$

Using this and (3.2),

$$\begin{aligned}\widehat{Z}(-t) &= W(-t)^*Z(-t)W(-t)\\ &= -W(t)^*Z(t)W(t)\\ &= -\widehat{Z}(t).\end{aligned}$$

Thus $\widehat{Z}(t) = tA_C(t^2)$ with A_C analytic and Hermitian. By Rellich's theorem, the eigenvalues of the one-parameter Hermitian analytic family $A_C(s)$ admit analytic branches. Comparing the finitely many Taylor germs lexicographically selects one branch $\alpha_*(s)$ that is maximal for all sufficiently small $s > 0$. For $t > 0$,

$$\lambda_{\max}(Z(t)) = \lambda_{\max}(\widehat{Z}(t)) = t\alpha_*(t^2).$$

Finally,

$$\log f(t) = \log \lambda_{\max}(Q(t)) = \lambda_{\max}(Z(t)),$$

which proves the theorem.

3.4 First matrix jet

Writing

$$C = C_H + C_A, \qquad C_A := \frac{C - C^*}{2}, \quad C_A^* = -C_A,$$

a Baker–Campbell–Hausdorff expansion of the canonical normal form gives

$$\boxed{A_C(s) = -2C_H + \frac{s}{12}[C_A, [C_H, C_A]] + O(s^2).}$$

For $x \in \ker(C_H)$,

$$\left\langle x, \frac{1}{12}[C_A, [C_H, C_A]]x \right\rangle = -\frac{1}{6}\langle C_A x, C_H C_A x\rangle \leq 0.$$

This recovers the $m = 1$ leading coefficient directly from ordinary degenerate Hermitian perturbation theory.

4 Answer to part (a)

Let

$$a := 2m + 1.$$

The known leading term and the odd logarithmic normal form imply

$$\kappa_0 = \cdots = \kappa_{m-1} = 0, \qquad \kappa_m = -c_1.$$

Hence

$$f(t) = \exp\Big(-c_1 t^a + \kappa_{m+1} t^{a+2} + \kappa_{m+2} t^{a+4} + \cdots\Big).$$

4.1 Genuinely hypocoercive case: $m \geq 1$

Here $a \geq 3$, so products from the exponential cannot contribute before order $2a$. There is no logarithmic term of degree $a + 1$, because $a + 1$ is even. Therefore

$$\boxed{[t^{2m+2}]\, \|e^{-Ct}\|_2^2 = 0.}$$

In the notation of part (a),

$$\boxed{c_2 = 0 \qquad (m \geq 1).}$$

Thus there is no nontrivial secondary integer index controlling this coefficient. The previous remainder can be sharpened by one order:

$$\boxed{\|e^{-Ct}\|_2^2 = 1 - c_1 t^{2m+1} + O(t^{2m+3}), \qquad m \geq 1.}$$

The first potentially new independent scalar is κ_{m+1}, at degree $2m + 3$, not degree $2m + 2$.

4.2 Coercive edge case: $m = 0$

Now

$$\log f(t) = -c_1 t + O(t^3).$$

Exponentiation yields

$$f(t) = 1 - c_1 t + \frac{c_1^2}{2} t^2 + O(t^3),$$

so

$$\boxed{c_2 = \frac{c_1^2}{2} = 2\lambda_{\min}(C_H)^2, \qquad m = 0.}$$

Again, no independent secondary index is needed.

5 Answer to part (b): the complete logarithmic spectral jet

A sequence of integers alone cannot determine numerical Taylor coefficients: even for

$$C_\varepsilon = \begin{pmatrix} 1 & -\varepsilon \\ \varepsilon & 0 \end{pmatrix}, \qquad \varepsilon \neq 0,$$

the HC index is always $m = 1$, whereas $c_1 = \varepsilon^2/6$ varies continuously. The correct complete invariant is therefore an integer support hierarchy paired with continuous spectral-jet amplitudes.

5.1 Definition

Define the **logarithmic hypocoercivity jet** by

$$\mathcal{J}(C) := (\kappa_0, \kappa_1, \kappa_2, \ldots),$$

where the κ_r are the Taylor coefficients of the right-maximal eigenvalue branch of $A_C(s)$. Define its nonzero support indices by

$$m_1 := m, \qquad m_{\ell+1} := \min\{r > m_\ell : \kappa_r \neq 0\},$$

whenever the set is nonempty. The complete higher-order data are the pairs

$$\boxed{(m_\ell, \kappa_{m_\ell}).}$$

Generically $m_2 = m + 1$, but symmetries or algebraic cancellations can create gaps. These support indices locate the genuinely new logarithmic invariants; the amplitudes are indispensable.

5.2 Exact coefficient formula

Write

$$f(t) = \sum_{n=0}^{\infty} a_n t^n, \qquad a_0 = 1.$$

Then

$$\boxed{a_n = \sum_{\substack{k_0,k_1,\ldots\geq 0\\ \sum_{r\geq 0}(2r+1)k_r=n}} \prod_{r\geq 0} \frac{\kappa_r^{k_r}}{k_r!}.}$$

Only finitely many k_r occur for each fixed n. Equivalently, from $f' = (\log f)' f$,

$$\boxed{a_0 = 1, \qquad a_n = \frac{1}{n} \sum_{\substack{r\geq 0\\ 2r+1\leq n}} (2r+1)\kappa_r\, a_{n-(2r+1)}.}$$

These identities determine every coefficient exactly once the logarithmic jet is known.

The sign convention in part (b) of the problem statement is not stable for higher orders: the coefficients need not all be positive. If that statement is written as

$$f(t) = 1 - \sum_{j\geq 1} c_j t^{2m+j},$$

then simply

$$c_j = -a_{2m+j}.$$

5.3 Immediate sparsity consequences for $m \geq 1$

Before nonlinear products can occur,

$$a_n = \begin{cases} \kappa_{(n-1)/2}, & n \text{ odd and } 2m+1 \leq n < 4m+2, \\ 0, & n \text{ even and } 2m+2 \leq n < 4m+2. \end{cases}$$

In particular,

$$\begin{aligned} f(t) = 1 - c_1 t^{2m+1} + \kappa_{m+1} t^{2m+3} + \kappa_{m+2} t^{2m+5} + \cdots \\ + \kappa_{2m} t^{4m+1} + \frac{c_1^2}{2} t^{4m+2} + O(t^{4m+3}). \end{aligned}$$

Thus every even degree $2m+2, 2m+4, \ldots, 4m$ vanishes, and the first forced even term occurs at degree $4m+2$, with coefficient $c_1^2/2$.

6 Exact finite-order computation of the jet

For any requested order R, the following is a finite algebraic procedure.

1. Expand $Q(t) = e^{-C^* t} e^{-Ct}$ through degree $2R+1$.
2. Expand $Z(t) = \operatorname{Log} Q(t)$ to the same degree.
3. Expand the polar unitary

$$U(t) = S(t)(S(t)^* S(t))^{-1/2}, \qquad S(t) = e^{C^* t}.$$

4. Form $W(t) = U(t)^{-1/2}$ and

$$\widehat{Z}(t) = W(t)^* Z(t) W(t) = t \sum_{r=0}^{R} A_r t^{2r} + O(t^{2R+3}).$$

5. Compute the right-maximal analytic eigenvalue branch of
$$A_C(s) = A_0 + A_1 s + \cdots + A_R s^R + O(s^{R+1}).$$
Rellich–Kato perturbation theory gives $\kappa_0, \ldots, \kappa_R$. Multiplicities are handled by analytic spectral projections or, equivalently, by a Feshbach/Schur-complement reduction on the currently maximizing eigenspace. One must not use a simple-eigenvector formula before this reduction.
6. Convert the κ_r to the coefficients a_n with the recurrence above.

All operations in steps 1–4 are standard convergent matrix power-series operations near the identity. Step 5 is a finite-dimensional Hermitian analytic eigenvalue problem, so it has ordinary power-series branches rather than Puiseux branches.

For a simple selected eigenvalue after the required block reductions, the usual recursion is explicit. If
$$A_C(s)v(s) = \alpha_*(s)v(s),$$
with $v_0^* v(s) = 1$, then, after fixing $v_0^* v_j = 0$ for $j \geq 1$,
$$\kappa_n = v_0^* \sum_{j=1}^{n} A_j v_{n-j},$$
and the complementary components v_n are obtained by applying the reduced resolvent of $A_0 - \kappa_0 I$. The same recursion is applied after each degenerate Kato reduction.

7 Verification

7.1 Exact symbolic check: an $m = 1$ family

For
$$C_\varepsilon = \begin{pmatrix} 1 & -\varepsilon \\ \varepsilon & 0 \end{pmatrix},$$
exact symbolic expansion gives
$$\|e^{-C_\varepsilon t}\|_2^2 = 1 - \frac{\varepsilon^2}{6}t^3 + \left(\frac{\varepsilon^2}{60} + \frac{\varepsilon^4}{120}\right)t^5 + \frac{\varepsilon^4}{72}t^6 + O(t^7),$$
$$\log \|e^{-C_\varepsilon t}\|_2^2 = -\frac{\varepsilon^2}{6}t^3 + \left(\frac{\varepsilon^2}{60} + \frac{\varepsilon^4}{120}\right)t^5 + O(t^7).$$
The t^4 coefficient is exactly zero, while the t^6 coefficient is
$$\frac{1}{2}\left(\frac{\varepsilon^2}{6}\right)^2,$$
as predicted.

At $\varepsilon = 3/10$, this is the matrix in Example 2.12 of Achleitner–Arnold–Carlen, and the exact expansion continues as
$$\|e^{-Ct}\|_2^2 = 1 - \frac{3}{200}t^3 + \frac{627}{400000}t^5 + \frac{9}{80000}t^6 - \frac{14983}{80000000}t^7 - \frac{1881}{80000000}t^8 + O(t^9),$$
$$\log \|e^{-Ct}\|_2^2 = -\frac{3}{200}t^3 + \frac{627}{400000}t^5 - \frac{14983}{80000000}t^7 + O(t^9).$$

7.2 Exact symbolic check: an $m = 2$ chain

Let

$$C = \begin{pmatrix} 1 & -2 & 0 \\ 2 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad C_H = \operatorname{diag}(1, 0, 0).$$

Here

$$\sum_{j=0}^{1} (C^*)^j C_H C^j$$

is singular, whereas the sum through $j = 2$ is positive definite (its determinant is 16), so $m = 2$. Exact characteristic-polynomial recursion gives

$$\|e^{-Ct}\|_2^2 = 1 - \frac{1}{90}t^5 - \frac{11}{7560}t^7 + \frac{7}{32400}t^9 + \frac{1}{16200}t^{10} + \frac{3779}{59875200}t^{11} + \frac{11}{680400}t^{12} + O(t^{13}).$$

The t^6 and t^8 coefficients vanish. Moreover,

$$\frac{1}{16200} = \frac{1}{2}\left(\frac{1}{90}\right)^2,$$

and the t^{12} logarithmic coefficient cancels because

$$\frac{11}{680400} = \left(-\frac{1}{90}\right)\left(-\frac{11}{7560}\right).$$

Thus the logarithm again has only odd powers.

7.3 Coercive edge check

Replacing the $m = 1$ family by $I + C_\varepsilon$ multiplies the squared norm by e^{-2t}. Hence

$$\log \|e^{-(I+C_\varepsilon)t}\|_2^2 = -2t - \frac{\varepsilon^2}{6}t^3 + O(t^5),$$

and

$$\|e^{-(I+C_\varepsilon)t}\|_2^2 = 1 - 2t + 2t^2 + O(t^3).$$

This verifies $c_2 = c_1^2/2$ when $m = 0$.

8 Adversarial audit

- **Branch multiplicity:** The proof does not assume a simple singular value at $t = 0$; the odd matrix normal form and Rellich branches handle full degeneracy.
- **Right side versus two-sided norm:** The odd analytic function constructed above agrees with the maximal branch for $t > 0$. It need not equal $\log\|e^{-Ct}\|_2^2$ for negative t; for negative t, ordering of the singular-value branches reverses. No false identity $f(-t) = 1/f(t)$ is used.
- **Norm convention:** The cited theorem states the coefficient for $\|P(t)\|_2$; the formula in this log includes the factor of two required for $\|P(t)\|_2^2$.
- **Coercive edge case:** $m = 0$ is exceptional because squaring/exponentiation contributes already at degree two.

- **Signs:** Higher Taylor coefficients are not sign-definite, so the all-minus notation in part (b) is only a naming convention.
- **Discrete insufficiency:** Integer indices locate vanishing orders but cannot determine amplitudes. The pairs $(m_\ell, \kappa_{m_\ell})$, or equivalently the full spectral jet, are the complete data.
- **Scope:** The analytic argument is finite-dimensional. Infinite-dimensional operator norms can fail to be analytic at zero, so extending this exact hierarchy to unbounded or general Hilbert-space generators requires separate hypotheses.
- **Machine checks:** Exact rational symbolic expansions were obtained for independent $m = 1$ and $m = 2$ examples; numerical random-matrix checks also found even logarithmic coefficients at numerical roundoff. These checks support but are not substituted for the algebraic proof.

9 Final status and remaining gap

9.1 Strongest verified result

Part (a) is completely resolved:

$$\boxed{c_2 = 0 \ (m \geq 1), \qquad c_2 = c_1^2/2 \ (m = 0).}$$

Part (b) has a complete structural and finite-order computational answer:

$$\boxed{\|e^{-Ct}\|_2^2 = \exp\left(\sum_{r\geq 0} \kappa_r t^{2r+1}\right),}$$

where the κ_r are obtained from the maximal analytic eigenvalue branch of the explicitly constructed Hermitian family $A_C(s)$. The coefficient partition formula and recurrence then produce all higher coefficients.

9.2 Residual limitation

No single closed scalar expression analogous to the leading-minimum formula is written here for every κ_r; at high order, degenerate Kato/Feshbach reductions naturally produce increasingly large noncommutative polynomial expressions. This is a simplification/representation gap, not a gap in finite-order computability or in the parity theorem.

9.3 Highest-value next action

Implement the matrix-series plus Kato/Feshbach recursion as a computer-algebra routine, then simplify the first two genuinely new jets κ_{m+1} and κ_{m+2} into invariant formulas for selected multiplicity patterns.